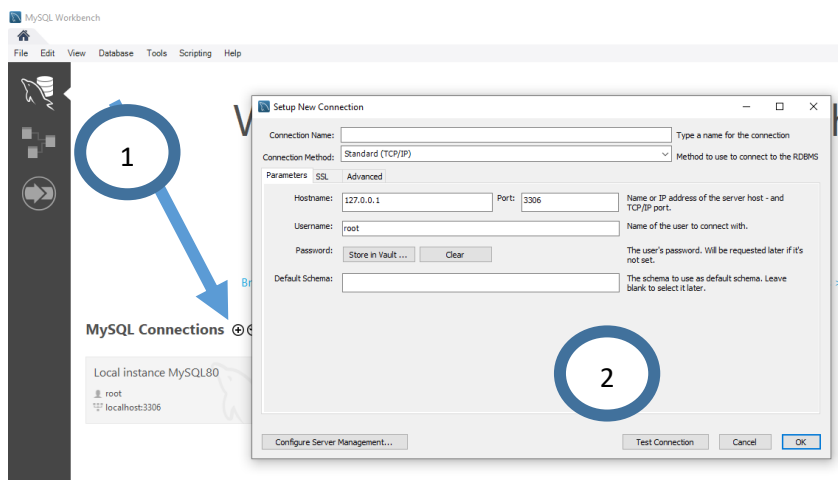


Lab 9 Information Systems

MySQL tutorial 1

In this lab, we will be using MySQL workbench to create a new database, insert data into it, perform a number of queries and use reverse engineering to create the ER diagram of the database.

STEP 1 Create a new database connection

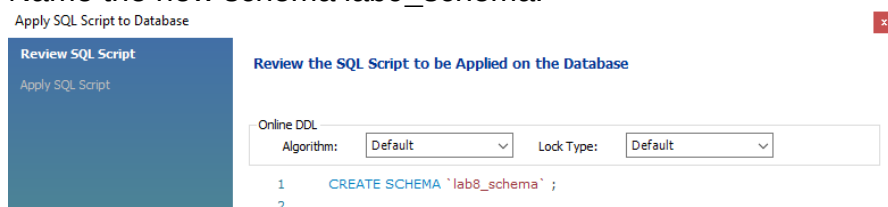


STEP 2 Create Database and tables

Once the connection is created, you may use that connection to enter SQL Editor to do SQL Development tasks like creating/modifying/deleting modify database and tables, running SQL queries etc.

Right click on any Schema and select Create Schema to create a new Schema.

Name the new schema lab9__schema:



Create table

Use GUI or the SQL statement below to create the following table:

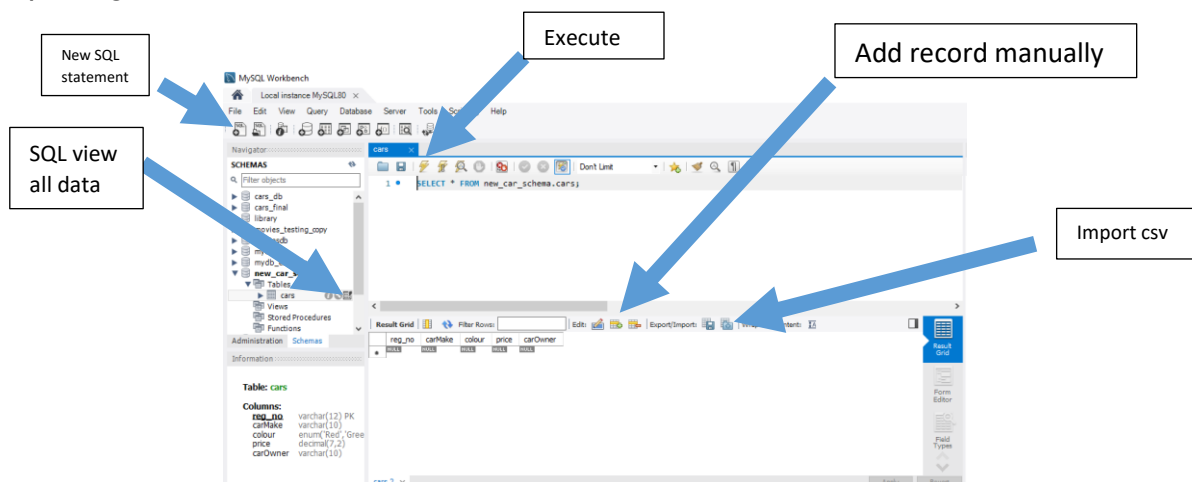
The screenshot shows the MySQL Workbench Table Designer for a table named 'cars' in the 'new_car_schema' schema. The table has five columns: 'reg_no' (VARCHAR(12), NOT NULL, PRIMARY KEY), 'carMake' (VARCHAR(10), NOT NULL), 'colour' (ENUM('Red', 'Green', 'Blue'), NOT NULL), 'price' (DECIMAL(7,2), NOT NULL), and 'carOwner' (VARCHAR(10), NULL). The 'reg_no' column is marked as the primary key.

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
reg_no	VARCHAR(12)	☒	☒	☐	☐	☐	☐	☐	☐	
carMake	VARCHAR(10)	☐	☒	☐	☐	☐	☐	☐	☐	
colour	ENUM('Red', 'Green', 'Blue')	☐	☒	☐	☐	☐	☐	☐	☐	
price	DECIMAL(7,2)	☐	☒	☐	☐	☐	☐	☐	☐	
carOwner	VARCHAR(10)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

```
CREATE TABLE cars (  
  reg_no VARCHAR(12) NOT NULL,  
  carMake VARCHAR(10) NOT NULL,  
  colour ENUM('Red', 'Green', 'Blue') NULL,  
  price DECIMAL(7,2) NOT NULL,  
  carOwner VARCHAR(10) NULL DEFAULT  
  NULL,  
  PRIMARY KEY (reg_no));
```

STEP 3 Insert table data

We can insert data in three different ways: (i) Manually, (ii) using SQL code and (iii) importing from a file.



- (i) Enter a car manually into your table
- (ii) Write an SQL statement to insert the following data:

regNo	carMake	colour	price	carOwner
201 WX 1278	Ford	Red	12000	Jim Smith
13 D 5671	Mercedes	Blue	22000	Bob Smith
14 MH 1278	Skoda	Blue	11000	Jim Smith
16 D 1124	Fiat	Green	9000	Bob Jones

Sample code

```
# insert 5 records into the "drivers" table  
INSERT INTO cars (regNo, carMake, colour, price, carOwner)  
VALUES ( '00 WX 1278', 'Ford', 'Red', '12000', 'Jim Smith' );
```

- (iii) Use the provided csv file (cars.csv) to import extra data into your table.

STEP 4 Perform Queries

We will perform basic queries as well as using aggregation functions such as: MAX , MIN , AVG, COUNT and SUM.

Examples:

```
SELECT * FROM cars
```

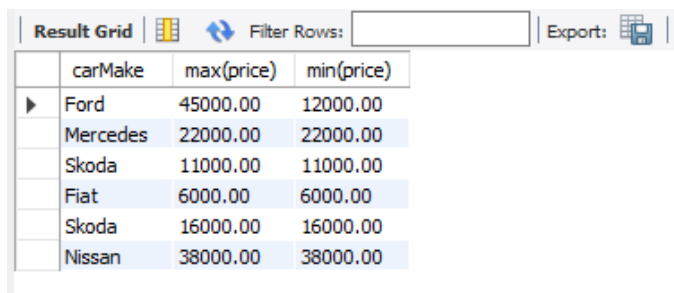
```
SELECT carMake, price FROM cars
```

```
SELECT carMake, price FROM cars WHERE carMake="Ford"
```

```
SELECT carMake, price, carOwner FROM cars WHERE carOwner LIKE "J%") ;
```

###GROUP BY and aggregation Functions

```
SELECT carMake, max(price), min(price) FROM cars GROUP BY carMake
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays the results of a query: 'SELECT carMake, max(price), min(price) FROM cars GROUP BY carMake'. The table has three columns: 'carMake', 'max(price)', and 'min(price)'. The data is grouped by car manufacturer. The interface also includes a 'Filter Rows' search bar and an 'Export' button.

carMake	max(price)	min(price)
Ford	45000.00	12000.00
Mercedes	22000.00	22000.00
Skoda	11000.00	11000.00
Fiat	6000.00	6000.00
Skoda	16000.00	16000.00
Nissan	38000.00	38000.00

```
SELECT carMake, COUNT(reg_No) AS Number_of_cars FROM cars GROUP BY carMake
```

###Using DISTINCT

```
SELECT COUNT(carMake) FROM cars
```

```
SELECT COUNT(DISTINCT carMake) FROM cars
```

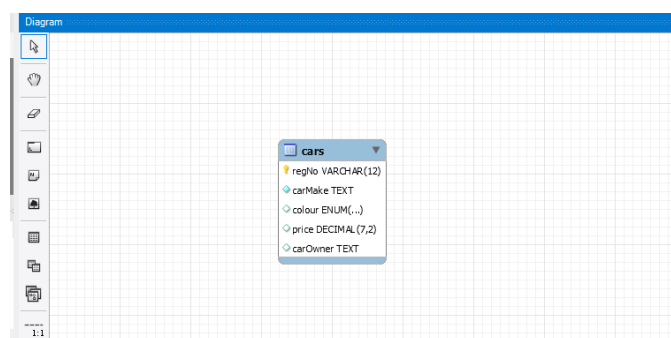
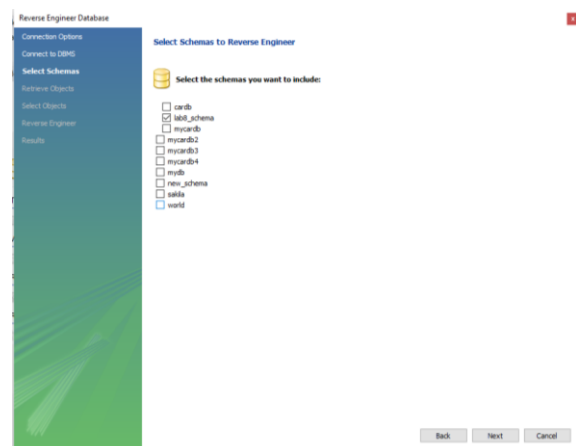
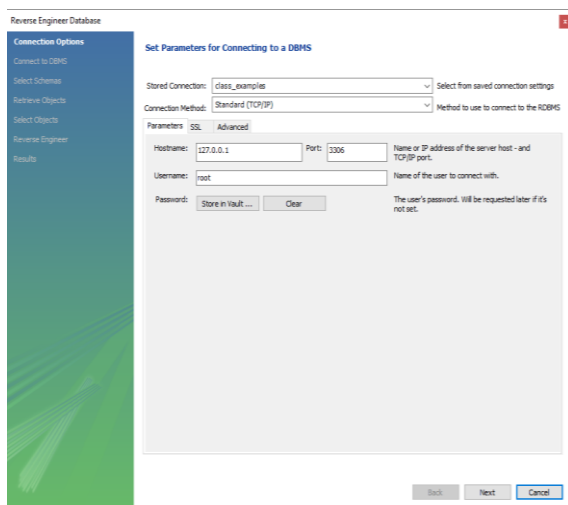
New queries

1. Create a query to retrieve all records from the table.
2. Create a query to retrieve all cars owned by Jim Smith.
3. Create a query to retrieve all blue cars.
4. Create a query to list all cars sorted by price ascending.
5. Create a query that calculates how many red cars are in stock.
6. Create a query that calculates the sum of the prices of all Ford cars.
7. Create a query that calculates the average price per colour.
8. Create a query that calculates the average, max and min price of all cars.
9. Create a query that calculates the max, average and min price of all cars by make.
10. Create a query that calculates the max, average and min price of all cars with a Dublin reg.

STEP 5 Data Modelling

You can create a new Entity Relationship model using various components available, creating ER diagram from an existing database using reverse engineering.

The following images show how to create an ER diagram from an existing database.



Use reverse engineering with your cars database as well as the sample database sakila.

References:

https://www.w3schools.com/sql/sql_count_avg_sum.asp

https://www.w3schools.com/sql/sql_min_max.asp

<https://www.w3resource.com/mysql/administration-tools/mysql-workbench-tutorial.php>